

科目 微積分 類組別 A2 共 2 頁 第 1 頁

一、填充題：共8題，每題8分、總計64分。請在答案卷上列出題號並依序作答。

請注意：本部分共8題，命題型態為填充題，請依題號順序獨立列出，勿同時陳列出計算過程，倘若答案被包含在演算過程內，將被視為試算流程，不另行挑出計分。答案必須完全正確才給分。

(1) Evaluate

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n+1} + \sqrt{n}}{n(\sqrt{n+1} - \sqrt{n})}.$$

(2) Define the function

$$F(x) = f(\ln(2 \sin x)), \quad 0 < x < \pi,$$

where f is differentiable on $(-\infty, \infty)$ and $f'(0) = 2$. Find $F'(\pi/6)$.

(3) Find the following integral

$$\int e^{x^3} x^2 dx.$$

(4) Find all critical points of the function

$$f(x, y) = x^3 - 6xy - y^2, \quad -\infty < x < \infty \text{ and } -\infty < y < \infty,$$

and classify each critical point as a location where a local maximum, local minimum, or saddle point occurs.

(5) Find the rational number that is represented by the repeating decimal

$$2.\overline{21} = 2.21212121\dots$$

(6) Calculate the area of the region that is completely enclosed by the graphs of the functions $f(x) = x^2$ and $g(x) = 4 - x^2$.

(7) Evaluate the double integral

$$\iint_R 20x^2 e^{-y} dA,$$

where $R = \{(x, y) : 0 \leq x \leq 3, 0 \leq y \leq 2\}$.

(8) Find the domain of the following function:

$$f(x) = \frac{\sqrt{4-x^2}}{\ln(x-1)}.$$

(後面還有題目)

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二、計算、證明題：共3題，每題12分、總計36分。請將題號標明清楚。

請注意：須詳細寫出計算及證明過程，否則不予計分。請書寫清楚乾淨，並將過程解釋清楚，任何書寫或是解釋不清將被扣分。

(1) Find the Taylor series of the function

$$f(x) = \frac{x}{3 + 2x}$$

and determine its radius of convergence.

(2) Use the separation of variables to solve the following differential equation:

$$\frac{dQ}{dt} = 2(100 - Q), \quad t \geq 0,$$

with the initial condition $Q(0) = 3$.

(3) To create an open box from a square piece of cardboard, we can cut out identical squares from each corner and then fold up the resulting flaps. If the cardboard measures 10 inches on each side, determine the dimensions of the box that will yield the maximum volume. Note: Points will be awarded only if calculus is used in the solution.

注意：背面有試題